Assignment #1: Introduction to Python for Machine Learning

Possible Points: 100

Deadline: 02/09/2026

Instructions: All assignments must be completed on an individual basis. Submit your Jupyter Notebook (`.ipynb`), the PDF (`.pdf`) file, and any other associated csv files if any.

Question 1: Sales Data Analysis (Pandas & Matplotlib)

(40 Points)

Scenario: You have a CSV file named `sales.csv` containing sales data.

1. Data Preparation (10 Points)

a. Import the data into a DataFrame.

b. Add a new column named `Total_Sale` which calculates the total sale of each product (`Price` * `Quantity`).

c. Calculate and print the **Mean**, **Median**, and **Standard Deviation** of this new column.

2. Visualization (15 Points)

a. Create a **bar graph** showcasing total sales data for each month using Matplotlib.

b. Create a **scatter plot** to check the correlation between `Quantity` and `Price`. Use the `corr()` function to print the correlation value. Briefly explain your findings in a markdown cell.

3. Filtering & Export (15 Points)

a. Filter the data to find rows where `Quantity > 10` **AND** the sale occurred after the month of May.

b. Save this filtered data into a new CSV file named `filtered_sales.csv`.

Question 2: Library Inventory Management (Pandas)

(30 Points)

Key	Title	Author	Year	Availability
Book1	The Great Gatsby	F. Scott Fitzgerald	1925	True
Book2	To Kill a Mockingbird	Harper Lee	1960	False
Book3	“1984”	George Orwell	1949	True
Book4	Pride and Prejudice	Jane Austen	1813	True
Book5	The Catcher in the Rye	J.D. Salinger	1951	True

Scenario: You are managing a library. Refer to the "Book Inventory" screenshot provided.

1. DataFrame Creation (10 Points)

a. Create a Pandas DataFrame that exactly matches the table in the screenshot (Book1 through Book5).

b. Ensure the columns are named: `Key`, `Title`, `Author`, `Year`, and `Availability`. Set `Key` as the index.

2. Updating Data (10 Points)

a. Imagine "To Kill a Mockingbird" (Book2) has just been returned to the library.

b. Write a script to update the `Availability` of Book2 from `False` to `True`. Print the updated DataFrame to show the change.

3. Search & Selection (10 Points)

a. Perform a filter operation to find all books written by the author "F. Scott Fitzgerald". Print the result.

b. Perform a filter operation to show only the books that are currently `True` for Availability. Print the result.

Question 3: Exam Score Analysis (NumPy)

(30 Points)

Maths	English	Biology	History
85	92	78	90
76	88	92	80
90	85	88	95
82	78	88	75
88	90	85	92

Scenario: Refer to the "Exam Scores" screenshot provided. You need to analyze the performance of 5 students across 4 subjects.

1. Matrix Creation (10 Points)

a. Create a 2D NumPy matrix called `exam_scores` containing the **exact data** shown in the screenshot. (The matrix should be 5 rows by 4 columns).

b. In a markdown cell, explain one key difference between a Python List and a NumPy Array.

2. Statistical Analysis (10 Points)

a. Use NumPy functions to calculate and print the **Maximum** and **Minimum** scores for **each subject** (Column-wise).

b. Calculate the average score for the entire class (all subjects, all students) and print it.

3. Advanced Analysis (10 Points)

a. Use `np.argmax` to find the **Student** (Row Index) who had the highest total score across all subjects.

b. Use `np.argmin` to find the **Subject** (Column Index) that had the lowest average score.